PART 1 — ENGLISH · 第一部分 — 英文

Request for Budgetary Technical and Commercial Proposal No. OVE75-RFQ-10

Subject: Evaporators, stages I and II (4 off), 904L.

KVANT LLC (Russian Federation), within the Basic Engineering (FEED) stage of the construction project of a copper production plant — a roasting–leaching–electrowinning complex with a capacity of 75,000 tonnes of copper cathodes per year — kindly requests your budgetary technical and commercial proposal for the item above. Project site: an operating metallurgical site in north-west Russia.

Item summary: Copper sulphate duty — D = 2.0 m (22,800 and 22,000 $m^3/(m^2 \cdot h)$); mixed sulphate duty — D = 1.3 m; t = 40 and 20 °C. Technical requirements are given in Annex 1 to this request.

Please include in your quotation:

1. budget price quoted separately on EXW (manufacturer’s works) and DDP project site, north-west Russia (Incoterms 2020), stating the currency and the validity period of the offer;
2. manufacturing lead time and estimated delivery time (weeks from the date of order / advance payment);
3. overall dimensions and weights of a unit and of the largest shipping blocks (for the logistics study);
4. reference list of similar supplies: product, process duty and parameters, year of supply, country;
5. materials of the parts in contact with the process medium, and the scope of supply (instrumentation, spare parts, documentation);
6. utility consumption and installation requirements (if available).

This is a budgetary enquiry at the Basic Engineering (FEED) stage: it does not constitute an offer and creates no obligations for either party; detailed technical data sheets will be issued at the next project stage.

Reply deadline: we kindly ask you to submit your quotation within 3 (three) weeks from receipt of this request to stepan@kvantpro.com. Please confirm receipt of this request.

Confidentiality. This request and the attached technical materials are provided solely for the preparation of the quotation, are confidential, and shall not be disclosed to third parties, copied or published without the prior written consent of KVANT LLC. Submission of a quotation creates no legal obligations for either party.

Contact person: Stepan Kharkov, KVANT LLC, stepan@kvantpro.com.

Annex 1: Technical requirements for the requested item.

Annex 1. Technical Requirements

To RFQ No. OVE75-RFQ-10 · The values follow the Customer's input data and terms of reference; Basic Engineering (FEED) stage; subject to refinement at the next project stage.

RFQ item	Evaporators, stages I and II (4 off), 904L
Summary	Copper sulphate: D = 2.0 m (22,800 and 22,000 $m^3/(m^2 \cdot h)$); mixed sulphate: D = 1.3 m; t = 40 and 20 °C
Specific requirements	The material of the wetted parts (904L) and the specific secondary-vapour load must be stated

Equipment covered by this request:

1. **Stage I evaporator (vacuum evaporator) — copper sulphate crystallization · tag 4.1.1**

Type / model	UUSS: EV-2 evaporator with external heater
Duty	Vacuum crystallization of recirculated copper sulphate, stage I
Key parameters	Specific vapour load 22,800 $m^3/(m^2 \cdot h)$ of secondary vapour; vapour flow 49,417 m^3/h; design D 2.17 m
Quantity	1 (1 train)
Materials	Steel 904L or equivalent
Dimensions, hold-up	D = 2.0 m; actual 3.1 m^2; margin factor 1.4
Additional requirements	Indirect (dead) steam heating, t = 40 °C

2. **Stage II evaporator (vacuum evaporator) — copper sulphate crystallization · tag 4.1.2**

Type / model	UUSS: EV-2 evaporator with forced circulation, operating at additional vacuum
Duty	Vacuum crystallization of copper sulphate, stage II
Key parameters	Specific vapour load 22,000 $m^3/(m^2 \cdot h)$; vapour flow 23,032 m^3/h; design D 1.05 m
Quantity	1
Materials	Steel 904L or equivalent
Dimensions, hold-up	D = 2.0 m; actual 3.1 m^2; margin factor 3.0
Additional requirements	t = 20 °C; operation by the deeper vacuum provided by the steam ejector

3. **Stage I evaporator — mixed sulphate crystallization · tag 4.1.23**

Duty	Vacuum crystallization of mixed copper–nickel sulphate, stage I, t = 40 °C

Key parameters	Specific vapour load 22,000 m³/(m²·h); vapour flow 17,647 m³/h; design D 0.80 m
Quantity	1
Materials	Steel 904L or equivalent
Dimensions, hold-up	D = 1.3 m; margin factor 1.7
Additional requirements	The evaporator diameter has been increased based on the refining calculations provided in the TR documentation

4. **Stage II evaporator — mixed sulphate crystallization · tag 4.1.24**

Duty	Vacuum crystallization of mixed sulphate, stage II, t = 20 °C
Key parameters	Specific vapour load 22,000 m³/(m²·h); vapour flow 5,530 m³/h; design D 0.25 m
Quantity	1
Materials	Steel 904L or equivalent
Dimensions, hold-up	D = 1.3 m; margin factor 5.3

The quotation must confirm: process medium, temperature, pressure, materials of wetted parts, capacity, dimensions and weight, power consumption, manufacturing lead time, delivery terms.

PART 2 — CHINESE · 第二部分 — 中文

预算性技术商务报价邀请函 编号 OVE75-RFQ-10

事由:一、二级蒸发器(4 台),材质 904L。

KVANT LLC(俄罗斯联邦)正在开展一座铜冶炼厂建设项目的基础设计(FEED)——"焙烧—浸出—电积"联合工艺装置,阴极铜产能为每年 75,000 吨。现请贵方就上述设备提交预算性技术商务报价。项目场地:俄罗斯西北部一座在运营的冶金厂区。

设备简要特性:硫酸铜工况——D = 2.0 m(22,800 及 22,000 $m^3/(m^2·h)$);混合硫酸盐工况——D = 1.3 m;温度 40 及 20 °C。 技术要求见本函附件 1。

请在报价中包含:

1. 预算价格——分别按 EXW(制造商仓库)及 DDP 项目场地(俄罗斯西北部)条件(Incoterms 2020)报价,并注明币种及报价有效期;
2. 制造周期及预计交货期(自订单/预付款之日起,按周计);
3. 单台设备及最大运输单元的外形尺寸和重量(用于物流研究);
4. 类似供货业绩表:产品、工艺介质及参数、供货年份、国家;
5. 与工艺介质接触部件的材质,以及供货范围(仪表、备件、技术文件);
6. 能耗及安装要求(如有资料)。

本询价为基础设计(FEED)阶段的预算性询价:不构成要约,对双方均不产生义务;详细技术数据表将在项目下一阶段发出。

回复期限:请于收到本函后 3(三)周内将报价发送至 stepan@kvantpro.com。请确认收到本函。

保密条款。本函及随附技术资料仅供编制报价使用,属保密信息,未经 KVANT LLC 事先书面同意,不得向第三方披露、复制或发布。提交报价不构成双方的法律义务。

联系人：Stepan Kharkov, KVANT LLC, stepan@kvantpro.com。

附件 1:本询价设备的技术要求。

(职务) (姓名) (签字)

附件 1. 技术要求

对应询价函编号 OVE75-RFQ-10 · 参数依据业主原始数据及设计任务书;基础设计(FEED)阶段,后续阶段可能进一步细化。

询价设备	一、二级蒸发器(4 台),材质 904L
简要特性	硫酸铜:D = 2.0 m(22,800 及 22,000 $m^3/(m^2·h)$);混合硫酸盐:D = 1.3 m;温度 40 及 20 °C
特殊要求	必须注明接触部件材质(904L)及二次蒸汽(汁汽)比负荷

本询价涵盖的设备清单:

1. **一级蒸发器(真空蒸发器)——硫酸铜结晶 · 位号 4.1.1**

型号	UUSS:配外置加热器的 EV-2 蒸发器
用途	循环硫酸铜真空结晶,一级
主要参数	二次蒸汽比负荷 22,800 $m^3/(m^2·h)$;蒸汽流量 49,417 m^3/h;计算直径 D 2.17 m
数量	1(1 条生产线)

材质	904L 钢或同等材质
尺寸、持液量	D = 2.0 m;实际 3.1 m^2;裕度系数 1.4
附加要求	间接(闭式)蒸汽加热,t = 40 °C

2. **二级蒸发器(真空蒸发器)——硫酸铜结晶 · 位号 4.1.2**

型号	UUSS:强制循环 EV-2 蒸发器,在附加真空下运行
用途	硫酸铜真空结晶,二级
主要参数	比负荷 22,000 $m^3/(m^2\cdot h)$;蒸汽流量 23,032 m^3/h;计算直径 D 1.05 m
数量	1
材质	904L 钢或同等材质
尺寸、持液量	D = 2.0 m;实际 3.1 m^2;裕度系数 3.0
附加要求	t = 20 °C;借助蒸汽喷射器提供的更深真空运行

3. **一级蒸发器——混合硫酸盐结晶 · 位号 4.1.23**

用途	铜镍混合硫酸盐真空结晶,一级,t = 40 °C
主要参数	比负荷 22,000 $m^3/(m^2\cdot h)$;蒸汽流量 17,647 m^3/h;计算直径 D 0.80 m
数量	1
材质	904L 钢或同等材质
尺寸、持液量	D = 1.3 m;裕度系数 1.7
附加要求	蒸发器直径已根据 TR 文件中提供的核算结果加大

4. **二级蒸发器——混合硫酸盐结晶 · 位号 4.1.24**

用途	混合硫酸盐真空结晶,二级,t = 20 °C
主要参数	比负荷 22,000 $m^3/(m^2\cdot h)$;蒸汽流量 5,530 m^3/h;计算直径 D 0.25 m
数量	1
材质	904L 钢或同等材质
尺寸、持液量	D = 1.3 m;裕度系数 5.3

报价中必须确认:工艺介质、温度、压力、接触部件材质、生产能力、尺寸与重量、能耗、制造周期、交货条件。